

Manual

Overview

`dt-csp-fairness-tool` finds Individual Discriminatory Instances (IDIs) in pre-trained neural network models. An IDI is a pair of inputs identical except for a sensitive feature (e.g., gender, race, age) where the model's output differs by more than 0.05 — indicating the model treats people differently based on a protected characteristic.

Three methods are implemented and compared against a random-search baseline. The baseline samples real data rows and flips **all** sensitive features simultaneously (e.g. sex, race, and age together), consistent with the reference implementation in `lab4_solution.py`. All three methods use the same simultaneous flip definition so comparisons are like-for-like.

Method	Description
<code>dt</code>	DT-only — random seed sampling to label pairs, trains a <code>DecisionTreeClassifier</code> to identify high-discrimination feature regions, then samples more inputs from those regions
<code>csp</code>	CSP-only — uses CP-SAT (Google OR-Tools) to generate inputs across the feature domain with diversity constraints to avoid revisiting known discriminatory regions
<code>hybrid</code>	Hybrid — Phase 1 (DT seed sampling + rule extraction) followed by Phase 2 (CSP constrained to DT-identified regions)

Primary finding: DT is the strongest method, consistently and substantially beating the baseline. CSP and Hybrid are included as ablation comparisons and provide valuable evaluation content around the trade-offs of constraint-based vs data-driven search.

Setup

1. Clone the repository
2. Install dependencies: `pip install -r requirements.txt`
3. Ensure `dataset/` and `DNN/` folders contain the required files (see below)

Windows note: Commands below use `python`. On some Windows setups the Python Launcher is invoked as `py` instead (e.g. `py main.py ...`). Use whichever works in your environment. On Linux/macOS `python3` may be required depending on your PATH.

Required Files

Datasets (`dataset/`):

Primary (required for main experiment):

- `processed_kdd.csv`
- `processed_adult.csv`
- `processed_compas.csv`
- `processed_german.csv`

- `processed_dutch.csv`
- `processed_credit_with_numerical.csv`

Ablation only (required for `--dataset all`):

- `processed_communities_crime.csv`
- `processed_law_school.csv`

Models (DNN/):

Primary:

- `model_processed_kdd_cleaned.h5`
- `model_processed_adult.h5`
- `model_processed_compas.h5`
- `model_processed_greman_cleaned.h5` ← note: typo in source filename, keep as-is
- `model_processed_dutch.h5`
- `model_processed_credit.h5`

Ablation only:

- `model_processed_communities_crime.h5`
- `model_processed_law_school_cleaned.h5`

Running the Tool

```
# Full experiment – all 8 datasets, all methods (recommended; expect several
hours)
python main.py --dataset all --method all --budget 1000 --runs 20

# Full experiment saving to a separate folder (e.g. to preserve a previous run)
python main.py --dataset all --method all --budget 1000 --runs 20 --results-dir
results_v2

# All three methods vs baseline on a single dataset (baseline shared, run once)
python main.py --dataset kdd --method all --budget 1000 --runs 20

# Multiple specific datasets (comma-separated, spaces optional)
python main.py --dataset kdd,adult,compas --method all --budget 1000 --runs 20

# Single method vs baseline
python main.py --dataset kdd --method dt      --budget 1000 --runs 20
python main.py --dataset kdd --method csp     --budget 1000 --runs 20
python main.py --dataset kdd --method hybrid --budget 1000 --runs 20

# Quick sanity check – 3 runs on one dataset to verify changes before a full run
python main.py --dataset adult --method all --runs 3 --results-dir results_test
```

Note: `--dataset all` runs all 8 datasets (KDD, Adult, COMPAS, Dutch, Credit, German, Law School, Communities & Crime). Law School and Communities & Crime are ablation-only datasets; to run only

the 6 primary datasets, either specify them individually or use a comma-separated list.

Arguments:

Argument	Default	Description
<code>--dataset</code>	required	Single dataset name, comma-separated list (e.g. <code>kdd, adult, compas</code>), or <code>all</code> (all 8 datasets)
<code>--method</code>	<code>hybrid</code>	<code>dt</code> , <code>csp</code> , <code>hybrid</code> , or <code>all</code>
<code>--budget</code>	<code>1000</code>	Total inputs generated per run
<code>--runs</code>	<code>20</code>	Independent trials per method (min 20 for meaningful Wilcoxon p-values)
<code>--seed-ratio</code>	<code>0.15</code>	Fraction of budget used for Phase 1 seed sampling (DT and Hybrid only)
<code>--results-dir</code>	<code>results</code>	Directory to write output CSVs — use a different path to avoid overwriting a previous run

Output

Results are saved incrementally — each run is written immediately so partial results survive an interrupted experiment. Output goes to `--results-dir` (default: `results/`):

- `<results-dir>/<dataset>_baseline.csv` — per-run IDI ratios for random search
- `<results-dir>/<dataset>_dt.csv` — per-run IDI ratios for DT-only
- `<results-dir>/<dataset>_csp.csv` — per-run IDI ratios for CSP-only
- `<results-dir>/<dataset>_hybrid.csv` — per-run IDI ratios for Hybrid
- `<results-dir>/summary.csv` — Wilcoxon comparison summary (means, std, p-values, timing); updated after each dataset completes when running `--dataset all`

IDI ratio = unique discriminatory inputs found / total inputs generated. Higher is better — a higher ratio means the method found more discrimination per unit of budget.

Configuration

Key parameters in `src/config.py`:

Parameter	Default	Description
<code>DEFAULT_BUDGET</code>	<code>1000</code>	Total inputs per run — imported by <code>main.py</code> as the <code>--budget</code> default
<code>DEFAULT_SEED_RATIO</code>	<code>0.15</code>	Phase 1 seed fraction — imported by <code>main.py</code> as the <code>--seed-ratio</code> default
<code>DT_MAX_DEPTH</code>	<code>4</code>	Reference value documenting the <code>max_depth</code> default in <code>dt.train_dt()</code> — not imported at runtime; change the default in <code>src/dt.py</code> directly to override
<code>DIVERSITY_K</code>	<code>1</code>	Reference value documenting the <code>k</code> default in <code>csp.add_diversity_constraint()</code> — not imported at runtime; change the argument in <code>src/csp.py</code> directly to override

Group Fairness Characterisation

A standalone script is provided to profile each model's group-level bias before running IDI search. It computes two metrics per sensitive feature:

- **Demographic Parity Difference (DPD)** — max difference in positive prediction rate across sensitive attribute groups. Does not require ground truth labels.
- **Equalized Odds Difference (EOD)** — max of TPR difference and FPR difference across groups. Requires the ground truth target column (already present in all dataset CSVs).

```
# Characterise all configured datasets
python scripts/characterise.py

# Single dataset
python scripts/characterise.py --dataset adult
```

Results are printed as a table per dataset and saved to `characterisation.csv` at the repo root. This script is intended as a one-time characterisation pass; re-running it produces the same output (no randomness involved).

Testing

The test suite covers all modules with 147 tests total. No real datasets or `.h5` model files are required — tests use `MockModel` fixtures and synthetic `pandas` DataFrames.

```
# Run all tests
python -m pytest tests/ -v

# Run per module
python -m pytest tests/test_dt.py -v # 33 tests
python -m pytest tests/test_csp.py -v # 47 tests
python -m pytest tests/test_hybrid.py -v # 24 tests
python -m pytest tests/test_evaluate.py -v # 31 tests
python -m pytest tests/test_baseline.py -v # 12 tests
```

`scipy` and `numpy` RuntimeWarnings (e.g. Wilcoxon on all-zero differences) are suppressed via `conftest.py` and do not indicate failures.
